

KinChat Structure Migration Plan

Scope: reorganize package/folder layout only. No behavior change, no schema change, no new features. Every command below is meant to be pasted directly into Termux from your project root (the directory containing `app/` and `settings.gradle.kts`).

1. Migration Analysis

What's already good (left untouched)

- `data/local/db/*` — the Room layer (23 files) is cohesive and correctly organized. No change.
- `domain/repository`, `domain/model`, `domain/usecase` — thin, shared contract layer used across features. Already correct as a flat, type-first layer. No change.
- `features/chat/info/*` and `features/chat/insights/*` — these already use a self-contained domain/data/di-per-feature pattern. This is the right model; nothing here should be flattened back into the app-root layers.
- `features/search/*` — same pattern, same verdict.
- `navigation/*`, `di/*` — appropriately scoped, descriptively named. Not restructured (see below).
- `data/source/auth/*` — a domain-scoped subfolder nested inside a layer-type folder (`data/source/`). This precedent is actually reused below for the chat data layer.

The real problems

1. `data/repository/` is a **24-file monolith**. Of those 24 files, 18 are chat-specific (`ChatRepositoryImpl`, sync engine, chat-settings handlers, operation handlers) sitting flat alongside 6 unrelated single-file repos for Auth, Dashboard, Contacts, Settings, Redis. Some chat files are partially nested (`settings/`, `mapper/`, `sync/models`, `sync/handlers/`), others aren't (`ChatSyncManager`, `PendingOperationWorker`, `SyncTrigger`, `PendingSyncCoordinator`, `ChatSessionProvider`, `ChatDataModels`, `ChatMessageManager` all sit loose at the top level). The inconsistency, not the nesting itself, is the problem.

2. **The push-notification pipeline is split across three unrelated-sounding packages** purely because of generic layer naming: `core/services/` (FCM listener, token manager, processor, wakelock), `core/services/models/` (payload DTO), `core/receivers/` (the notification action receiver), and 9 more notification files buried in the `core/Utils/` grab-bag (builders, channel/summary managers, action + shortcut helpers). For a messaging app this is one of the most complex, most-touched subsystems in the codebase — it deserves one boundary, not three.
3. **Two logging utilities are stranded outside `core/logging/`**. `DebugLogger.kt` and `CrashLogManager.kt` live in `core/Utils/` while `LogMessage`, `LogRepository`, `AppLogger` (the rest of the same subsystem) live in `core/logging/`.
4. **`ui/theme/*` sits at the package root**, as a sibling to `core/`, while `core/ui/` also exists. Two differently-named `ui` packages at different depths is confusing for anyone navigating the tree. Design tokens are core infrastructure and should live under `core/`.

Explicit decisions on ambiguous cases (per your naming-collision rule)

- `data/repository/sync/handlers/AuthOperationHandler.kt` — despite the "Auth" name, this stays with its sibling `MessageOperationHandler` / `ReactionOperationHandler` / `ParticipantOperationHandler`. It's almost certainly one of several handlers implementing the same interface consumed by `PendingOperationDispatcher` — i.e. it's part of the sync engine, not the auth feature. Verify before moving anything (command included in Phase 4).
- `data/remote/api/ChatNotificationService.kt` / `ChatRpcService.kt` — NOT moved into the new `core/notifications/` package despite the name. These are remote-communication interfaces (calling a server endpoint to trigger a push), not the on-device notification-building pipeline. They correctly stay under `data/remote/api/`, consistent with how `AuthApi` is organized there.
- `Countries.kt`, `MediaDownloader.kt`, `ChatFormatters.kt`, `BatteryOptimizationHelper.kt` — left in `core/Utils/`. Each is either genuinely used by 2+ features or its usage is unverified from the file list alone. `Countries.kt` gets a conditional phase (5) with a grep check before any move.
- `di/*` — deliberately left flat. 13 well-named Hilt modules isn't a real problem, and moving DI wiring without full visibility into each module's `@InstallIn` / binding graph is a real risk for close to zero benefit. Not touched.

2. Migration Plan

Phase	What	Files touched	Risk
1	Git branch + baseline check	0	none
2	Design system + logging consolidation	10	low
3	Notification/push subsystem consolidation	15	medium — AndroidManifest references
4	Chat data-layer consolidation under <code>data/repository/chat/</code>	18	medium — touches sync engine, run unit tests
5	<code>Countries.kt</code> relocation (conditional on grep result)	0 – 1	low
—	Final verification & commit	—	—

3. Phase 1 — Git Branch & Baseline

Commands

```
git status
git checkout -b refactor/project-structure
```

Also save the helper script every later phase depends on:

```
mkdir -p scripts
cat > scripts/kinchat-repackage.sh << 'EOF'
#!/usr/bin/env bash
# Sourced helper for the KinChat structure migration.
# move_file "old/relative/Path.kt" "new/relative/Path.kt" ->
physically moves the file
# fix_refs "old/relative/Path.kt" "new/relative/Path.kt" -> fixes
its package line + every FQN reference to it

BASE="app/src/main/java/com/kinchat/app"
SRC_ROOT="app/src/main/java"
```

```

move_file() {
    local old="$1" new="$2"
    mkdir -p "$BASE/$(dirname "$new")"
    git mv "$BASE/$old" "$BASE/$new"
}

fix_refs() {
    local old="$1" new="$2"
    local class old_pkg new_pkg old_pkg_esc matches
    class=$(basename "$new" .kt)
    old_pkg="com.kinchat.app.$(dirname "$old" | tr '/' '.')"
    new_pkg="com.kinchat.app.$(dirname "$new" | tr '/' '.')"
    old_pkg_esc=$(printf '%s' "$old_pkg" | sed 's/\./\\./g')

    # fix this file's own package declaration
    sed -i "s/^package ${old_pkg_esc}\\$/package ${new_pkg}/"
"$BASE/$new"

    # fix every other file that references this exact class by fully-
    # qualified name
    matches=$(grep -r1 "${old_pkg_esc}\\.${class}" "$SRC_ROOT" 2>/dev/null)
    if [ -n "$matches" ]; then
        echo "$matches" | xargs sed -i
"s/${old_pkg_esc}\\.${class}/${new_pkg}.${class}/g"
    fi
}
EOF

```

Verification

```

git branch --show-current # -> refactor/project-structure
git status                 # clean working tree before proceeding

```

4. Phase 2 — Design System & Logging Consolidation

```

ui/theme/* → core/designsystem/*.
core/utils/{DebugLogger,CrashLogManager}.kt → core/logging/.

```

Commands

```

source scripts/kinchat-repackage.sh

move_file "ui/theme/Color.kt" "core/designsystem/Color.kt"
move_file "ui/theme/Theme.kt" "core/designsystem/Theme.kt"

```

```

move_file "ui/theme/Type.kt" "core/designsystem/Type.kt"
move_file "ui/theme/Typography.kt"
"core/designsystem/Typography.kt"
move_file "ui/theme/Shapes.kt" "core/designsystem/Shapes.kt"
move_file "ui/theme/Dimensions.kt"
"core/designsystem/Dimensions.kt"
move_file "ui/theme/Elevation.kt" "core/designsystem/Elevation.kt"
move_file "ui/theme/Animation.kt" "core/designsystem/Animation.kt"

move_file "core/utils/DebugLogger.kt"
"core/logging/DebugLogger.kt"
move_file "core/utils/CrashLogManager.kt"
"core/logging/CrashLogManager.kt"

rmdir app/src/main/java/com/kinchat/app/ui/theme
app/src/main/java/com/kinchat/app/ui 2>/dev/null

```

Package/Import Updates

```

source scripts/kinchat-repackage.sh

fix_refs "ui/theme/Color.kt" "core/designsystem/Color.kt"
fix_refs "ui/theme/Theme.kt" "core/designsystem/Theme.kt"
fix_refs "ui/theme/Type.kt" "core/designsystem/Type.kt"
fix_refs "ui/theme/Typography.kt" "core/designsystem/Typography.kt"
fix_refs "ui/theme/Shapes.kt" "core/designsystem/Shapes.kt"
fix_refs "ui/theme/Dimensions.kt" "core/designsystem/Dimensions.kt"
fix_refs "ui/theme/Elevation.kt" "core/designsystem/Elevation.kt"
fix_refs "ui/theme/Animation.kt" "core/designsystem/Animation.kt"

fix_refs "core/utils/DebugLogger.kt"
"core/logging/DebugLogger.kt"
fix_refs "core/utils/CrashLogManager.kt"
"core/logging/CrashLogManager.kt"

```

This also fixes `MainActivity.kt`'s theme import and any file that referenced `DebugLogger` / `CrashLogManager` automatically — the grep pass scans the whole source tree, not just the moved files.

Verification

```

grep -rn "com.kinchat.app.ui.theme" app/src/main/java
# expect nothing
grep -rn
"com.kinchat.app.core.utils.DebugLogger\|com.kinchat.app.core.utils.C

```

```
rashLogManager" app/src/main/java # expect nothing
./gradlew assembleDebug
```

5. Phase 3 — Notification/Push Subsystem Consolidation

Everything from `core/services/`, `core/services/models/`, `core/receivers/`, plus the 9 notification-named files in `core/utils/`, becomes one subsystem: `core/notifications/{fcm, builder, actions}/`.

Commands

```
source scripts/kinchat-repackage.sh

move_file "core/services/KinChatMessagingService.kt"
"core/notifications/fcm/KinChatMessagingService.kt"
move_file "core/services/FcmTokenManager.kt"
"core/notifications/fcm/FcmTokenManager.kt"
move_file "core/services/FcmMessageProcessor.kt"
"core/notifications/fcm/FcmMessageProcessor.kt"
move_file "core/services/WakeLockManager.kt"
"core/notifications/fcm/WakeLockManager.kt"
move_file "core/services/models/FcmMessagePayload.kt"
"core/notifications/fcm/model/FcmMessagePayload.kt"

move_file "core/utils/NotificationHelper.kt"
"core/notifications/builder/NotificationHelper.kt"
move_file "core/utils/NotificationStyleBuilder.kt"
"core/notifications/builder/NotificationStyleBuilder.kt"
move_file "core/utils/NotificationSummaryManager.kt"
"core/notifications/builder/NotificationSummaryManager.kt"
move_file "core/utils/NotificationChannelManager.kt"
"core/notifications/builder/NotificationChannelManager.kt"
move_file "core/utils/NotificationImageHelper.kt"
"core/notifications/builder/NotificationImageHelper.kt"
move_file "core/utils/NotificationTimeFormatter.kt"
"core/notifications/builder/NotificationTimeFormatter.kt"
move_file "core/utils/NotificationConstants.kt"
"core/notifications/builder/NotificationConstants.kt"

move_file "core/utils/NotificationActionHelper.kt"
"core/notifications/actions/NotificationActionHelper.kt"
move_file "core/utils/NotificationShortcutHelper.kt"
"core/notifications/actions/NotificationShortcutHelper.kt"
move_file "core/receivers/NotificationActionReceiver.kt"
```

```
"core/notifications/actions/NotificationActionReceiver.kt"

rmdir app/src/main/java/com/kinchat/app/core/services/models \
      app/src/main/java/com/kinchat/app/core/services \
      app/src/main/java/com/kinchat/app/core/receivers 2>/dev/null
```

Package/Import Updates

```
source scripts/kinchat-repackage.sh

fix_refs "core/services/KinChatMessagingService.kt"
"core/notifications/fcm/KinChatMessagingService.kt"
fix_refs "core/services/FcmTokenManager.kt"
"core/notifications/fcm/FcmTokenManager.kt"
fix_refs "core/services/FcmMessageProcessor.kt"
"core/notifications/fcm/FcmMessageProcessor.kt"
fix_refs "core/services/WakeLockManager.kt"
"core/notifications/fcm/WakeLockManager.kt"
fix_refs "core/services/models/FcmMessagePayload.kt"
"core/notifications/fcm/model/FcmMessagePayload.kt"

fix_refs "core/utils/NotificationHelper.kt"
"core/notifications/builder/NotificationHelper.kt"
fix_refs "core/utils/NotificationStyleBuilder.kt"
"core/notifications/builder/NotificationStyleBuilder.kt"
fix_refs "core/utils/NotificationSummaryManager.kt"
"core/notifications/builder/NotificationSummaryManager.kt"
fix_refs "core/utils/NotificationChannelManager.kt"
"core/notifications/builder/NotificationChannelManager.kt"
fix_refs "core/utils/NotificationImageHelper.kt"
"core/notifications/builder/NotificationImageHelper.kt"
fix_refs "core/utils/NotificationTimeFormatter.kt"
"core/notifications/builder/NotificationTimeFormatter.kt"
fix_refs "core/utils/NotificationConstants.kt"
"core/notifications/builder/NotificationConstants.kt"

fix_refs "core/utils/NotificationActionHelper.kt"
"core/notifications/actions/NotificationActionHelper.kt"
fix_refs "core/utils/NotificationShortcutHelper.kt"
"core/notifications/actions/NotificationShortcutHelper.kt"
fix_refs "core/receivers/NotificationActionReceiver.kt"
"core/notifications/actions/NotificationActionReceiver.kt"
```

This also fixes the imports inside `di/ChatServicesModule.kt` / `di/ChatManagersModule.kt` (or wherever these are provided to Hilt) automatically, since it's a plain import statement.

AndroidManifest Updates — do this manually, don't blind-sed

`KinChatMessagingService` (a `<service>`) and `NotificationActionReceiver` (a `<receiver>`) are almost certainly declared in the manifest. Check the actual notation first:

```
grep -n "KinChatMessagingService\|NotificationActionReceiver"
app/src/main/AndroidManifest.xml
```

Then apply **only the variant that matches what you saw**:

```
# If relative dot-notation, e.g.
android:name=".core.services.KinChatMessagingService"
sed -i
's/\.\core\.\services\.\KinChatMessagingService/\core.notifications.fcm.
KinChatMessagingService/' app/src/main/AndroidManifest.xml
sed -i
's/\.\core\.\receivers\.\NotificationActionReceiver/\core.notifications.
actions.NotificationActionReceiver/' app/src/main/AndroidManifest.xml
```

```
# If fully-qualified instead, e.g.
android:name="com.kinchat.app.core.services.KinChatMessagingService"
sed -i
's/com\.\kinchat\.\app\.\core\.\services\.\KinChatMessagingService/com.kin
chat.app.core.notifications.fcm.KinChatMessagingService/'
app/src/main/AndroidManifest.xml
sed -i
's/com\.\kinchat\.\app\.\core\.\receivers\.\NotificationActionReceiver/com
.kinchat.app.core.notifications.actions.NotificationActionReceiver/'
app/src/main/AndroidManifest.xml
```

Re-run the `grep` afterward to confirm both lines now point at the new packages.

Verification

```
grep -rn
"com.kinchat.app.core.services\|com.kinchat.app.core.receivers"
app/src/main/java app/src/main/AndroidManifest.xml # expect nothing
./gradlew assembleDebug
```

6. Phase 4 — Chat Data-Layer Consolidation

Extracts the 18 chat-specific files out of `data/repository/` into `data/repository/chat/`, leaving only the 5 genuinely single-domain repos (`AuthRepositoryImpl`, `DashboardRepositoryImpl`, `ContactsRepositoryImpl`, `SettingsRepositoryImpl`, `RedisRepositoryImpl`) at the top level.

Special handling — verify before moving

```
grep -n "class AuthOperationHandler" -A 3
app/src/main/java/com/kinchat/app/data/repository/sync/handlers/AuthOperationHandler.kt
```

Confirm it implements the same handler interface as

`MessageOperationHandler` / `ReactionOperationHandler` / `ParticipantOperationHandler`

before proceeding — if it does, it moves with its siblings below despite the "Auth" name (it's about auth-triggered *sync operations*, not the auth feature itself).

Commands

```
source scripts/kinchat-repackage.sh

move_file "data/repository/ChatRepositoryImpl.kt"
"data/repository/chat/ChatRepositoryImpl.kt"
move_file "data/repository/ChatDataModels.kt"
"data/repository/chat/ChatDataModels.kt"
move_file "data/repository/ChatSessionProvider.kt"
"data/repository/chat/ChatSessionProvider.kt"
move_file "data/repository/ChatMessageManager.kt"
"data/repository/chat/ChatMessageManager.kt"

move_file "data/repository/ChatSyncManager.kt"
"data/repository/chat/sync/ChatSyncManager.kt"
move_file "data/repository/PendingOperationWorker.kt"
"data/repository/chat/sync/PendingOperationWorker.kt"
move_file "data/repository/SyncTrigger.kt"
"data/repository/chat/sync/SyncTrigger.kt"
move_file "data/repository/PendingSyncCoordinator.kt"
"data/repository/chat/sync/PendingSyncCoordinator.kt"

move_file "data/repository/mapper/ChatSyncMapper.kt"
"data/repository/chat/sync/mapper/ChatSyncMapper.kt"
move_file "data/repository/sync/models/WorkerSyncDtos.kt"
"data/repository/chat/sync/models/WorkerSyncDtos.kt"
move_file "data/repository/sync/handlers/MessageOperationHandler.kt"
"data/repository/chat/sync/handlers/MessageOperationHandler.kt"
move_file "data/repository/sync/handlers/ReactionOperationHandler.kt"
```

```

"data/repository/chat/sync/handlers/ReactionOperationHandler.kt"
move_file
"data/repository/sync/handlers/ParticipantOperationHandler.kt"
"data/repository/chat/sync/handlers/ParticipantOperationHandler.kt"
move_file "data/repository/sync/handlers/AuthOperationHandler.kt"
"data/repository/chat/sync/handlers/AuthOperationHandler.kt"
move_file
"data/repository/sync/handlers/PendingOperationDispatcher.kt"
"data/repository/chat/sync/handlers/PendingOperationDispatcher.kt"

move_file "data/repository/settings/ChatSettingsConstants.kt"
"data/repository/chat/settings/ChatSettingsConstants.kt"
move_file "data/repository/settings/ChatLocalSettingsHandler.kt"
"data/repository/chat/settings/ChatLocalSettingsHandler.kt"
move_file "data/repository/settings/ChatRemoteSettingsHandler.kt"
"data/repository/chat/settings/ChatRemoteSettingsHandler.kt"

rmdir app/src/main/java/com/kinchat/app/data/repository/settings \
    app/src/main/java/com/kinchat/app/data/repository/mapper \
    app/src/main/java/com/kinchat/app/data/repository/sync/models \
    app/src/main/java/com/kinchat/app/data/repository/sync/handlers \
    app/src/main/java/com/kinchat/app/data/repository/sync
2>/dev/null

```

Package/Import Updates

```

source scripts/kinchat-repackage.sh

fix_refs "data/repository/ChatRepositoryImpl.kt"
"data/repository/chat/ChatRepositoryImpl.kt"
fix_refs "data/repository/ChatDataModels.kt"
"data/repository/chat/ChatDataModels.kt"
fix_refs "data/repository/ChatSessionProvider.kt"
"data/repository/chat/ChatSessionProvider.kt"
fix_refs "data/repository/ChatMessageManager.kt"
"data/repository/chat/ChatMessageManager.kt"

fix_refs "data/repository/ChatSyncManager.kt"
"data/repository/chat/sync/ChatSyncManager.kt"
fix_refs "data/repository/PendingOperationWorker.kt"
"data/repository/chat/sync/PendingOperationWorker.kt"
fix_refs "data/repository/SyncTrigger.kt"
"data/repository/chat/sync/SyncTrigger.kt"
fix_refs "data/repository/PendingSyncCoordinator.kt"
"data/repository/chat/sync/PendingSyncCoordinator.kt"

fix_refs "data/repository/mapper/ChatSyncMapper.kt"

```

```

"data/repository/chat/sync/mapper/ChatSyncMapper.kt"
fix_refs "data/repository/sync/models/WorkerSyncDtos.kt"
"data/repository/chat/sync/models/WorkerSyncDtos.kt"
fix_refs "data/repository/sync/handlers/MessageOperationHandler.kt"
"data/repository/chat/sync/handlers/MessageOperationHandler.kt"
fix_refs "data/repository/sync/handlers/ReactionOperationHandler.kt"
"data/repository/chat/sync/handlers/ReactionOperationHandler.kt"
fix_refs
"data/repository/sync/handlers/ParticipantOperationHandler.kt"
"data/repository/chat/sync/handlers/ParticipantOperationHandler.kt"
fix_refs "data/repository/sync/handlers/AuthOperationHandler.kt"
"data/repository/chat/sync/handlers/AuthOperationHandler.kt"
fix_refs
"data/repository/sync/handlers/PendingOperationDispatcher.kt"
"data/repository/chat/sync/handlers/PendingOperationDispatcher.kt"

fix_refs "data/repository/settings/ChatSettingsConstants.kt"
"data/repository/chat/settings/ChatSettingsConstants.kt"
fix_refs "data/repository/settings/ChatLocalSettingsHandler.kt"
"data/repository/chat/settings/ChatLocalSettingsHandler.kt"
fix_refs "data/repository/settings/ChatRemoteSettingsHandler.kt"
"data/repository/chat/settings/ChatRemoteSettingsHandler.kt"

```

Verification

```

find app/src/main/java/com/kinchat/app/data/repository -maxdepth 1 -
type f -name "*.kt"
# expect exactly: AuthRepositoryImpl.kt DashboardRepositoryImpl.kt
ContactsRepositoryImpl.kt SettingsRepositoryImpl.kt
RedisRepositoryImpl.kt

grep -rn
"com.kinchat.app.data.repository.settings\|com.kinchat.app.data.repos
itory.mapper\|com.kinchat.app.data.repository.sync" app/src/main/java
# expect nothing

./gradlew assembleDebug

```

This phase touches your offline-queue sync classes directly (`ChatSyncManager` , `PendingOperationWorker` , `PendingSyncCoordinator`) — run your existing unit test suite for that queue explicitly, not just the debug build:

```

./gradlew testDebugUnitTest

```

7. Phase 5 — `Countries.kt` (Conditional)

Only proceed if the grep confirms it's auth-exclusive.

```
grep -rln "core.utils.Countries" app/src/main/java
```

- If every hit is under `features/auth/`, move it:

```
source scripts/kinchat-repackage.sh
move_file "core/utils/Countries.kt"
"features/auth/utils/Countries.kt"
fix_refs "core/utils/Countries.kt"
"features/auth/utils/Countries.kt"
./gradlew assembleDebug
```

- If it's used anywhere else (contacts, settings, etc.), leave it in `core/utils/` — no action needed.

8. Final Verification & Commit

```
# 1. No stale references to any removed package
for pkg in \
  "com.kinchat.app.ui.theme" \
  "com.kinchat.app.core.services" \
  "com.kinchat.app.core.receivers" \
  "com.kinchat.app.data.repository.settings" \
  "com.kinchat.app.data.repository.mapper" \
  "com.kinchat.app.data.repository.sync"
do
  echo "checking: $pkg"
  grep -rn "$pkg" app/src/main/java app/src/main/AndroidManifest.xml
done

# 2. No duplicate class filenames left behind anywhere
find app/src/main/java -name "*.kt" -exec basename {} \; | sort |
uniq -d

# 3. No leftover empty directories
find app/src/main/java/com/kinchat/app -type d -empty

# 4. Full clean build (this alone will surface any broken Hilt
binding, Room reference, or Nav reference)
./gradlew clean assembleDebug

# 5. Full unit test suite (covers your offline-queue tests
```

```
specifically)
./gradlew testDebugUnitTest

# 6. Review the full diff before merging
git status
git diff --stat main...refactor/project-structure

# 7. Commit
git add -A
git commit -m "refactor: reorganize package structure (design system,
logging, notifications, chat data layer)"
```

Merge into `main` only after `assembleDebug` and `testDebugUnitTest` both pass clean on the branch, and you've manually smoke-tested push notifications on-device (since that's the phase with the manifest change).